PROJECT PROBLEM STATEMENT

Transformer and Equipment Condition Monitoring Service

This is a data-driven service for continuous health assessment, initial fault detection, and predictive maintenance of power transformers and associated substation equipment.

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Transformers: The backbone of the Power Grid

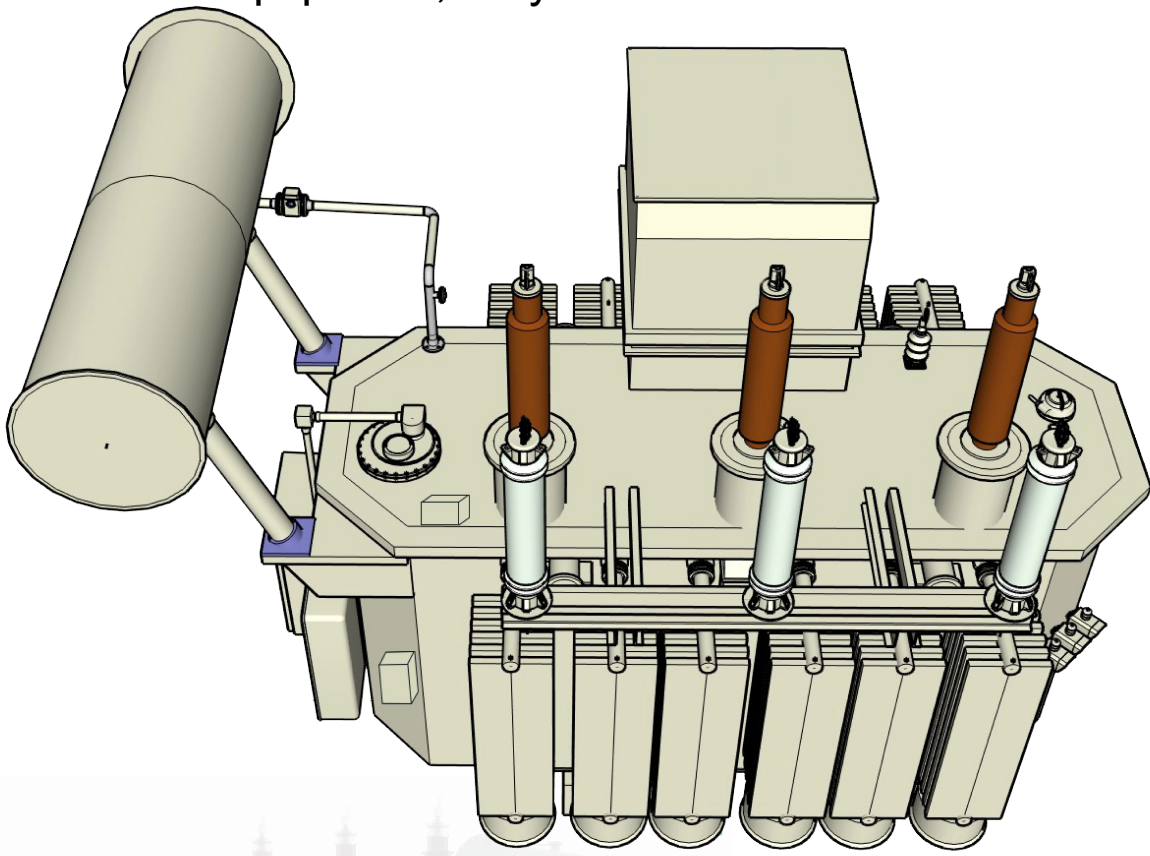
Power transformers are the **single most expensive and critical components** of electrical substations and power generating stations, where, along with their associated electrical equipment, they are critical to generation, transformation, protection, and transmission.

Function. Transformers step voltage up and down across generation, transmission, and distribution, with every unit of energy delivered passing through at least one of them.

Design life vs. real life. Units are engineered for 25-40 years of service, but a large share of the installed fleet is operating beyond design age, under rising load and switching stress.

Exposure. They are exposed to thermal ageing of oil-paper insulation, moisture, transient over-voltages, and short-circuiting. These mechanical forces accumulate silently over decades of operation.

Failure rate. Industry failure surveys covering report an annual transformer failure rate of about 0.5% - 1.0% - small per unit, severe per event.



Oil-immersed power transformer

One asset class, three utility settings

Generation, transmission and distribution each depend on transformers, but serve different parts of the power chain.

01

KenGen

Generation step-up

Steps up generator voltage for efficient evacuation.

→

02

KETRACO

Transmission substations

Moves bulk power across the high-voltage network.

→

03

Kenya Power (KPLC)

Distribution & customer supply

Supplies homes, businesses, public services and industry.



Konza substation

People exposed to service interruption

- Households and small enterprises
- Hospitals, water services and other essential facilities
- Industrial and commercial customers
- Utility field crews responding to faults

HOW FAILURES DEVELOP

A fault is often a process, not a surprise

Electrical, thermal, mechanical and environmental stresses can create detectable changes before an outage.

Stress
Overload • heat • moisture
contamination • ageing

→

Defect
Winding / insulation
weakness • loose joint

→

Signal
Temperature • gas • PD
load imbalance • oil change

→

Escalation
Insulation breakdown
protection trip / damage



After a critical failure: crews must isolate, repair or replace equipment.

Typical fault families

- Winding / insulation: short circuits, partial discharge, ageing
- Bushings: contamination, cracking, moisture ingress or dielectric failure
- Cooling / oil: blocked cooling, leaks, oxidation or overheating
- OLTC / connections: contact wear, arcing, loose joints and abnormal heating

The impact travels beyond the failed asset

A transformer outage can disrupt customers and essential services while increasing repair, replacement and network-restoration costs.

9.42 h

FY 2024/25

Average annual outage duration per customer

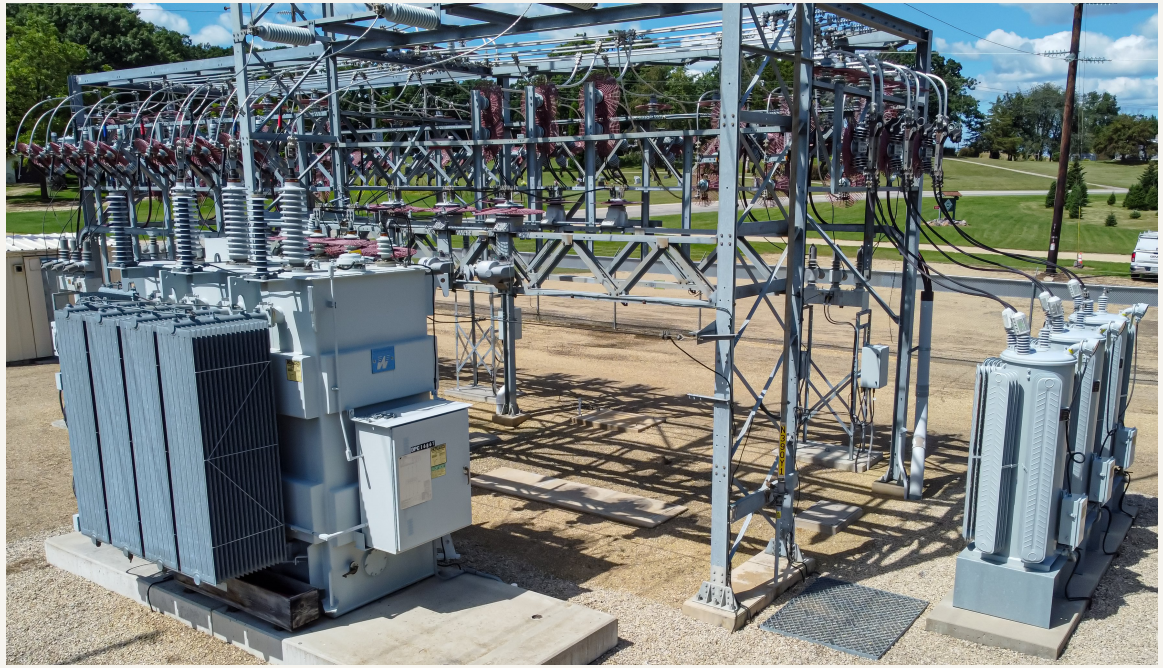
2.57 h

FY 2024/25

Average duration per customer interruption

System-wide reliability context

These national indicators describe customer interruptions overall; they do not attribute outages to transformer faults.



Who bears the consequences?

Customers • essential services • utility crews • network operators

WHAT CAN BE OBSERVED

Combine signals; no sensor tells the whole story

Existing protection, tests and inspections can be joined with selected online sensors where the asset and pilot justify them.

Thermal & loading	Top-oil / winding temperature, ambient temperature, current, loading and cooling state
Oil condition	Dissolved gas analysis (DGA), moisture, dielectric / oil quality and oil level
Electrical condition	Partial discharge, bushing condition, relay / trip events and power quality
Mechanical & maintenance	OLTC operations, vibration, inspection observations, tests and work orders



Thermography can reveal abnormal hot spots



Oil sampling adds laboratory condition evidence

KENYAN EVIDENCE

Monitoring capability is on the agenda; implementation status needs verification

Public documents point to activity, but a procurement, draft plan or tender specification does not prove live fleet-wide monitoring.

KETRACO • 2023 draft plan

Discusses automation and a remote sensing / asset-management pilot targeted for 2026.

Kenya Power • 2025/26 tender

A published procurement list includes a power-transformer partial-discharge monitoring system.

KenGen • 2023 transformer tender

Includes transformer monitoring and workstation capabilities in its technical requirements. Requirements alone do not establish what was commissioned.



The research opportunity

Verify asset data pathways, alarm ownership and maintenance response at a selected pilot site.

CORE PROBLEM

Signals can exist while a developing fault remains operationally invisible

The research tests where a warning is lost between measurement, interpretation, prioritisation and maintenance action.

1 Defect → **2 Signal** → **3 Decision** → **4 Action**

incipient change → sensor / test → interpretation → work order

Missing / delayed data	Offline tests are periodic; online sensors may be absent or disconnected.
No shared asset context	Measurements, alarm history and maintenance notes may sit in separate systems.
Threshold-only alerting	Static limits can miss asset-specific trends or create nuisance alarms.
Unclear ownership	A warning may lack a ranked action, assigned person and feedback loop.

Pilot hypothesis to verify

Fragmentation + weak trend-to-action workflow = avoidable late detection

PROPOSED SOFTWARE SERVICE

A human-centred workflow from condition evidence to maintenance action

The service complements protection relays and engineering judgment; it does not operate or trip the transformer.

01

Collect

Sensors • SCADA
lab tests • inspections
relay events • work orders

→

02

Validate

Asset identity
units • timestamps
quality • missing data

→

03

Interpret

Trends + rules
limits • rate of change
asset context

→

04

Prioritise

Explainable risk
severity • evidence
recommended check

→

05

Act & learn

Engineer review
assigned work order
feedback / outcome

Example alert

“Bushing phase B is heating faster than its own 30-day baseline under comparable load; inspect and confirm thermography.”

Every alert should show the asset, evidence, confidence / data quality and next review step.

HOW IMPROVEMENT WILL BE SHOWN

Evaluate the monitoring service, not just the software interface

Use a bounded dataset or pilot workflow and compare it with the documented current process.

Project deliverables

- Current-state process map and verified requirements
- Working prototype: asset register, data ingestion, trends, alerts and action log
- Explainable rule set and alert rationale
- Pilot / replay evaluation with test cases and user feedback
- Deployment, cybersecurity, data-governance and scale-up recommendations

Measures to report

Data completeness	Are expected measurements present and linked to the right asset?
Detection quality	Sensitivity / precision on available labelled or replay cases
Alert burden	False alarms per asset-month; duplicates and nuisance rate
Response workflow	Time to acknowledge / assign and completeness of action records
User assessment	Usefulness, clarity and trust in displayed evidence

Targets should be set after baseline measurement and stakeholder review.

THE PROBLEM TO SOLVE

How can condition evidence become an earlier maintenance decision?

In Kenya's electricity sector, transformers and associated equipment are exposed to electrical, thermal, mechanical and environmental stresses that can progress into severe faults and customer interruptions. Monitoring evidence may be distributed across online sensors, periodic tests, inspections, protection events and maintenance records. Without reliable asset linkage, trend interpretation, prioritised alerts and a closed action-feedback loop, developing defects can remain undetected or be acted on late.

RQ1 What data and workflow gaps delay condition-based action at the selected site?

RQ2 How can explainable rules and trends combine heterogeneous condition evidence?

RQ3 Does the prototype improve detection, alert quality or response workflow on the pilot cases?

Scope: one agreed site • decision support • prototype + evaluation

SELECTED SOURCES

Evidence and photo credits

Use the presenter notes throughout for claim-specific URLs and full image attribution.

Kenyan sector and technical sources

- EPRA. Statistics Report, June 2025 (FY 2024/25 reliability indicators).
- KETRACO. Climate Investment Funds Draft Plan, 10 Oct 2023.
- Kenya Power. Tender KP19A3OT2425-26; monitoring procurement listing.
- KenGen. Tender KGN-GDD-016-2023; transformer monitoring requirements.
- IEC 60599:2022. Mineral-oil equipment DGA interpretation guidance.
- IEEE C57.104-2019. Interpretation of gases generated in oil-immersed transformers.
- CIGRE TB 642. Transformer Reliability Survey.
- Samimi & Tenbohlen (2020). Online transformer condition monitoring review.

Embedded real-world photos

- Transformer at substation — Wikideas1, Wikimedia Commons, CC0
- Lineworker replacing transformer — Dave Pape, Wikimedia Commons, public domain
- Konza substation — The Star / Radio Africa
- Thermography of bushings — Reinhausen
- Transformer oil sampling — Oncor